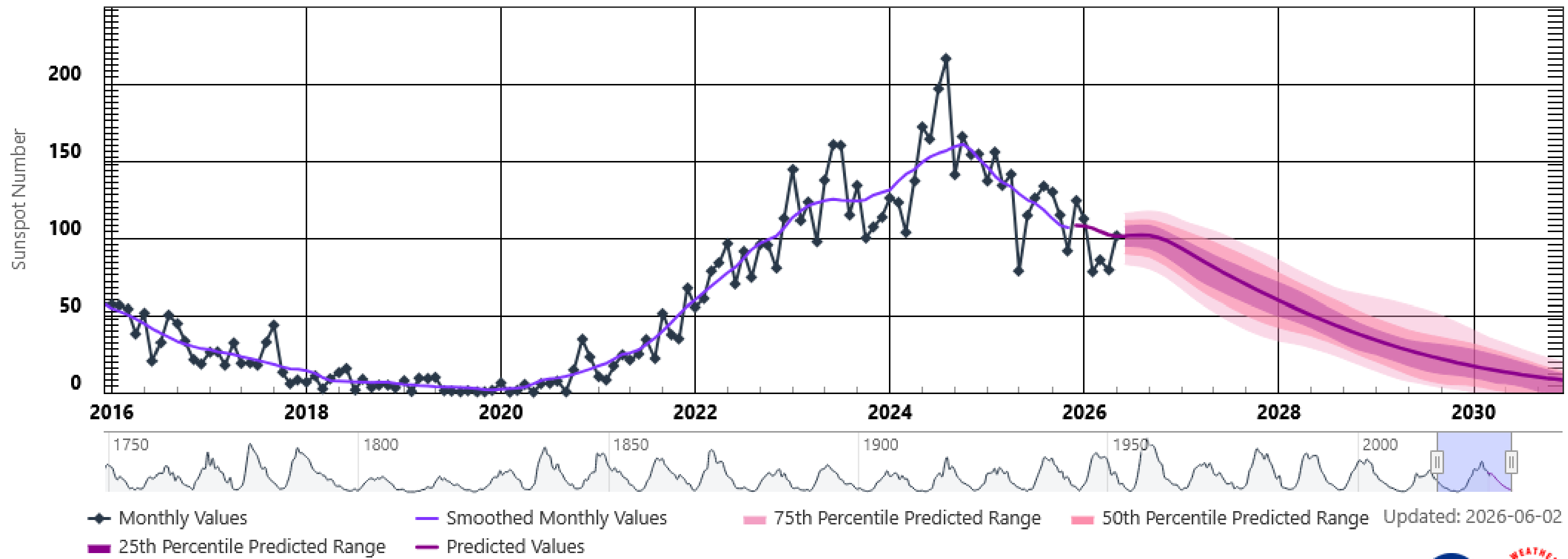
SOLAR WEATHER
2 JUN 2026

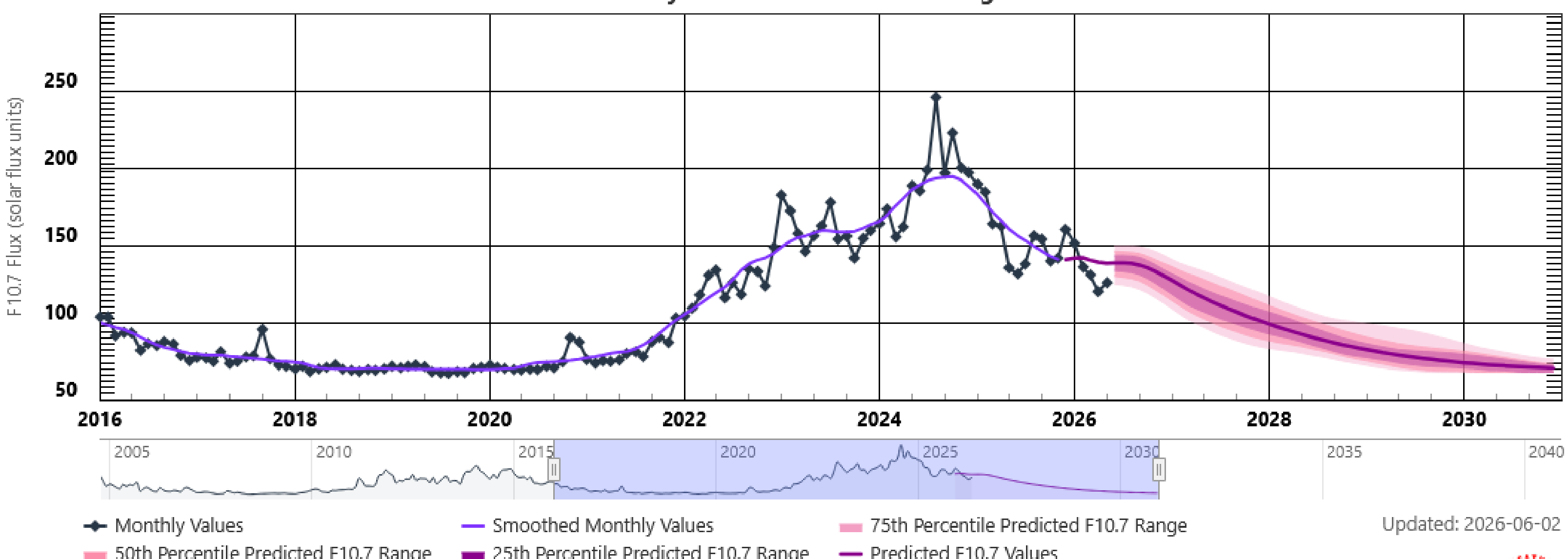
Lewis Thompson
W5IFQ

Taken by Dallas Poulson on January 20, 2026 @ Lake Louise (Banff), Alberta, Canada

Solar Cycle Sunspot Number Progression



Solar Cycle F10.7cm Radio Flux Progression



SolarHam

Indices: (6/2 @ 00:35 UTC) SFI 132 ▼ 4 SSN 146

3 Day Geomagnetic Forecast

June 2	June 3	June 4
2-3 (G0)	3-4 (G0)	3-4 (G0)

Max Kp

June 2	June 3	June 4
M-Lat 05%	M-Lat 10%	M-Lat 15%
H-Lat 30%	H-Lat 30%	H-Lat 50%

Probabilities

Latest SWPC Forecast (@ 00:30 + 12:30 UTC)

Detailed Forecast

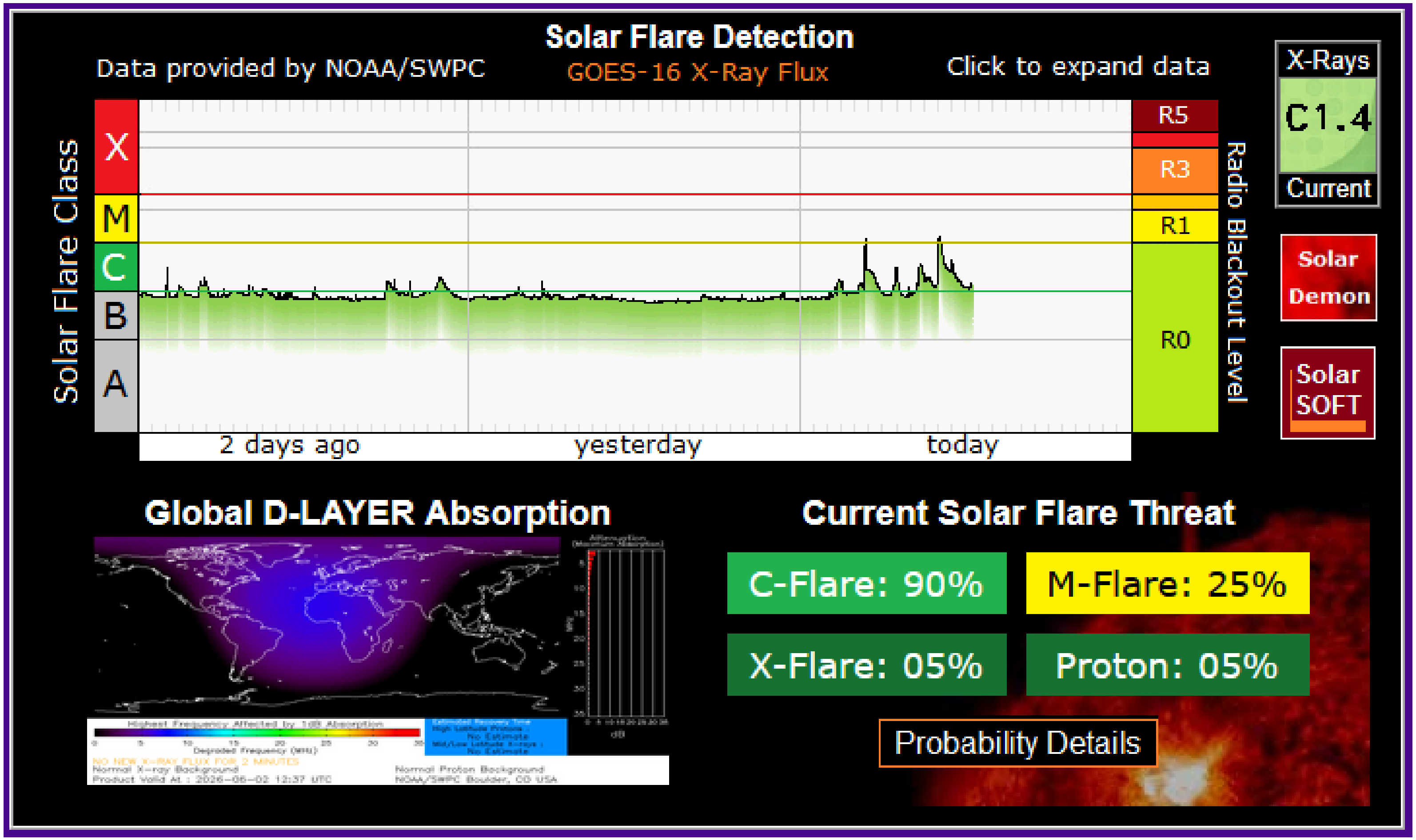
Visible Sunspot Regions

Sunspot Summary ▶ SRS

AR 4459	B	N14E63	*Growing*
AR 4458	B	S04E20	*Growing*
AR 4457	B	S09W27	*Declining*
AR 4455	B	N15E11	*Stable*
AR 4454	A	S15W07	*Declining*
AR 4449	A	S09W43	*Declining*
AR 4447	A	S17W82	*Stable*
AR 4446	B	S15W48	*Stable*
AR 4444	A	S22W73	*Declining*

Updated @ 00:45 UTC (June 2)

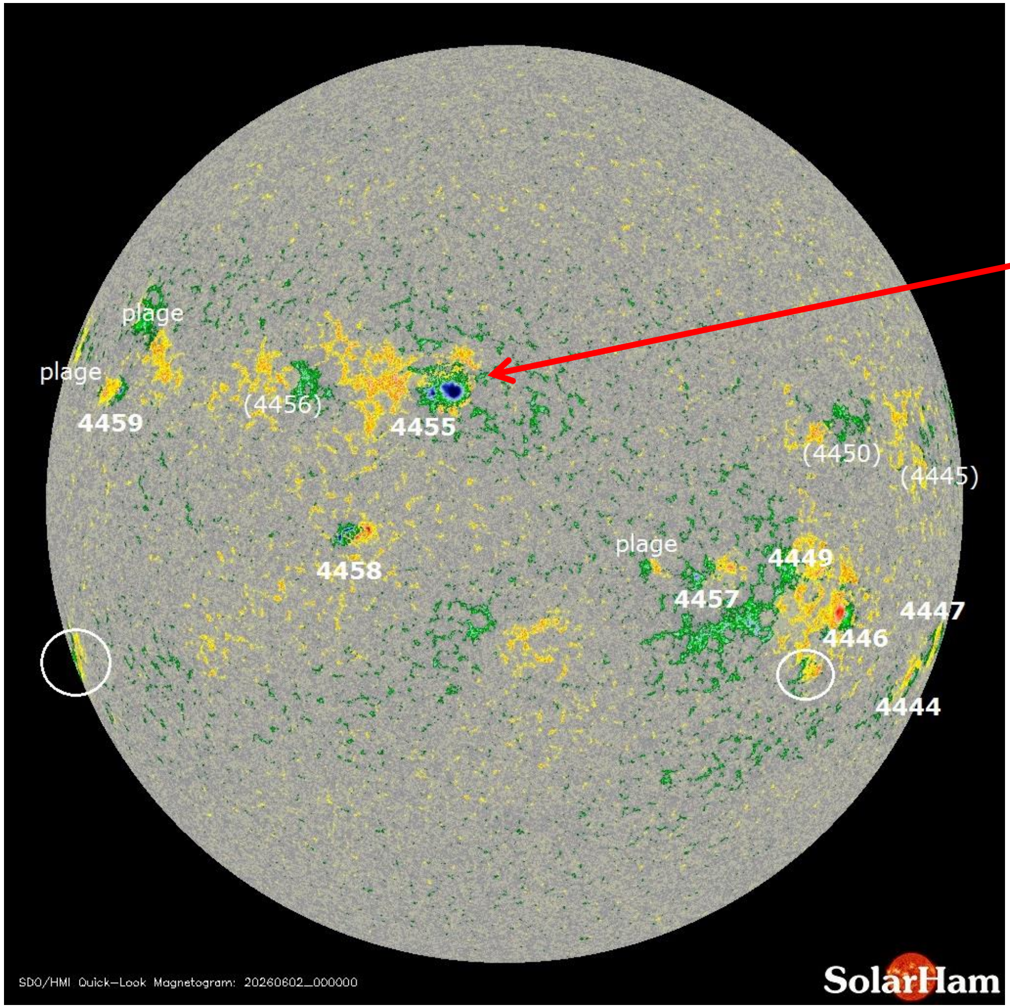
SolarHam

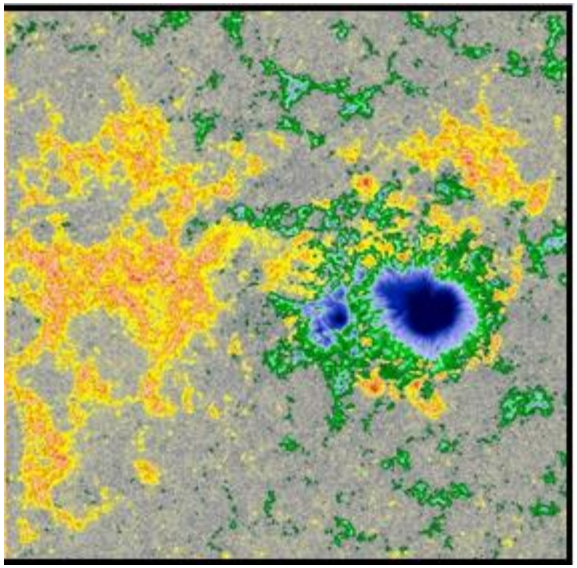


Solar Flare Detection
Data provided by NOAA/SWPC
GOES-16 X-Ray Flux
Click to expand data
X-Rays
C1.4
Current
Solar Flare Class
X
M
C
B
A
R5
R3
R1
R0
Radio Blackout Level
2 days ago
yesterday
today
Solar Demon
Solar SOFT
Global D-LAYER Absorption
Current Solar Flare Threat
C-Flare: 90%
M-Flare: 25%
X-Flare: 05%
Proton: 05%
Probability Details

Sun Spots

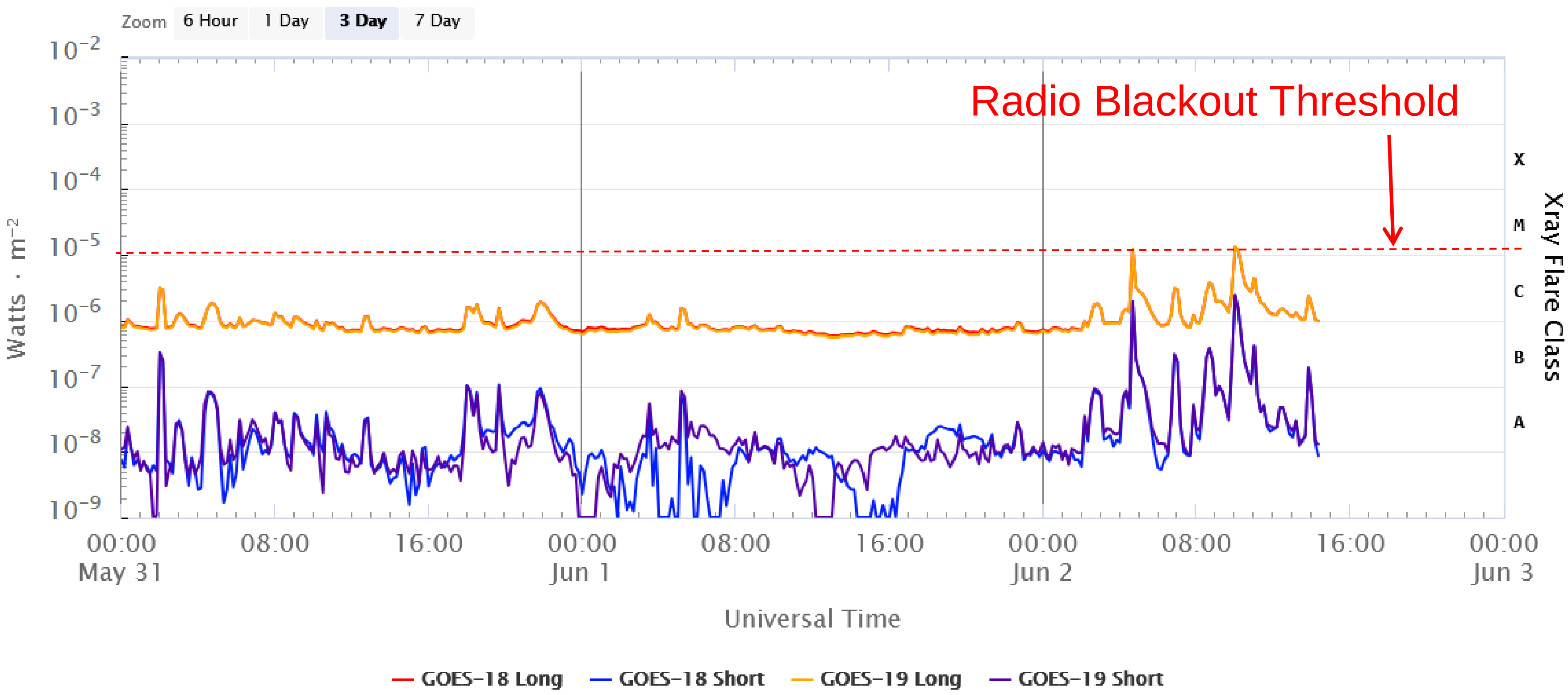
Magnetogram Image (Updated June 2, 2026)



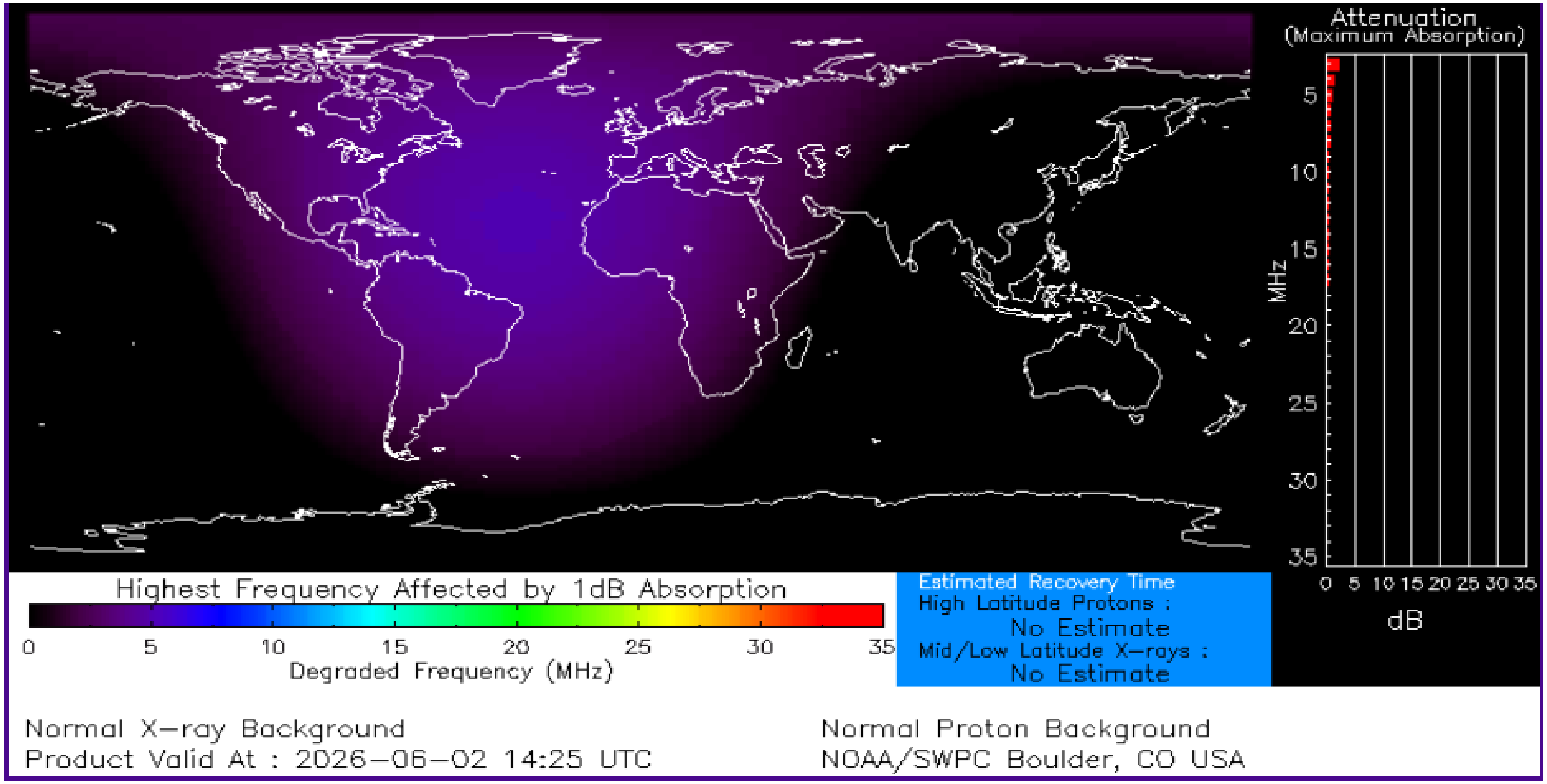


Beta
Produced M1.3 flare

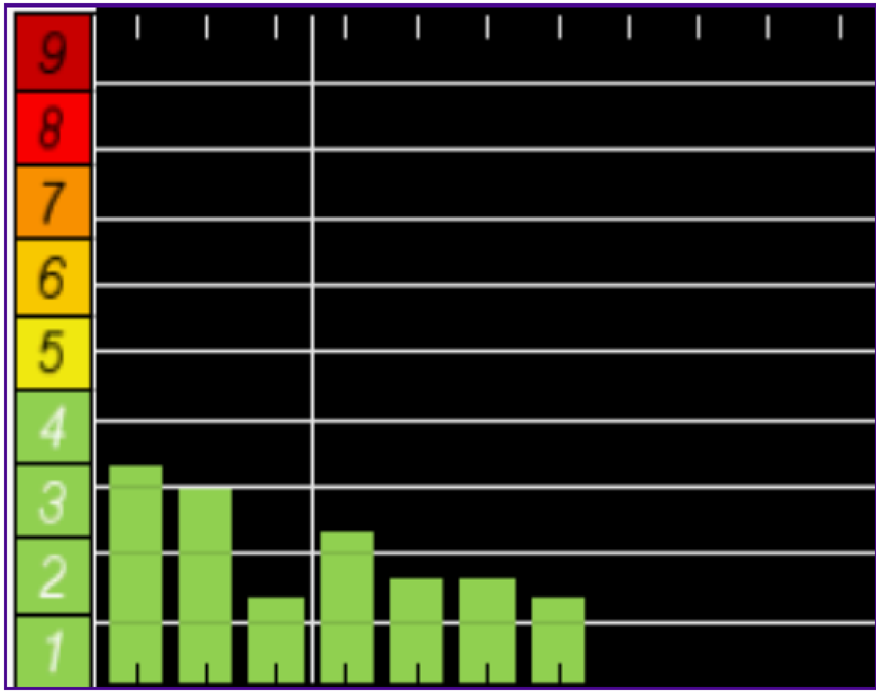
Solar X-Ray Flux: MAY 31 – 2 JUN 2026

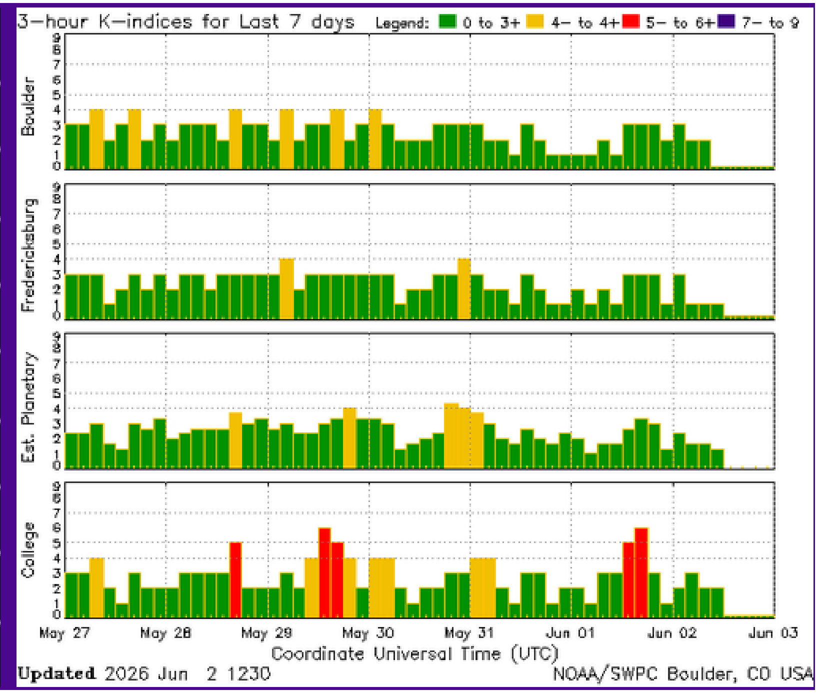


NOAA – D-Region Absorption Predictions



Earth's Geomagnetic Activity
2 – JUN 2026

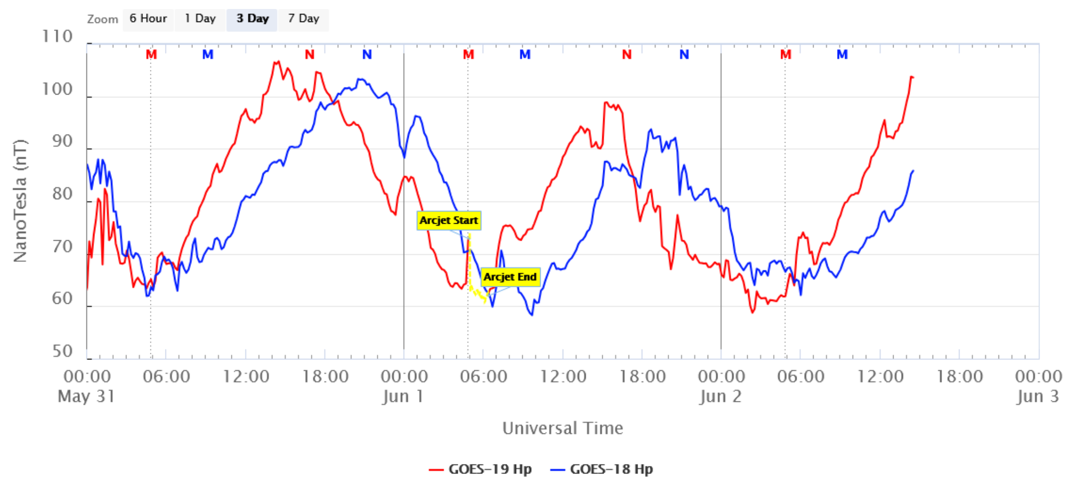




Geomagnetic Conditions: 2 JUN 2026

Solar wind:
Bz = 1.07 nT
speed = 383 km/sec
density = 3.89 protons/cm^3

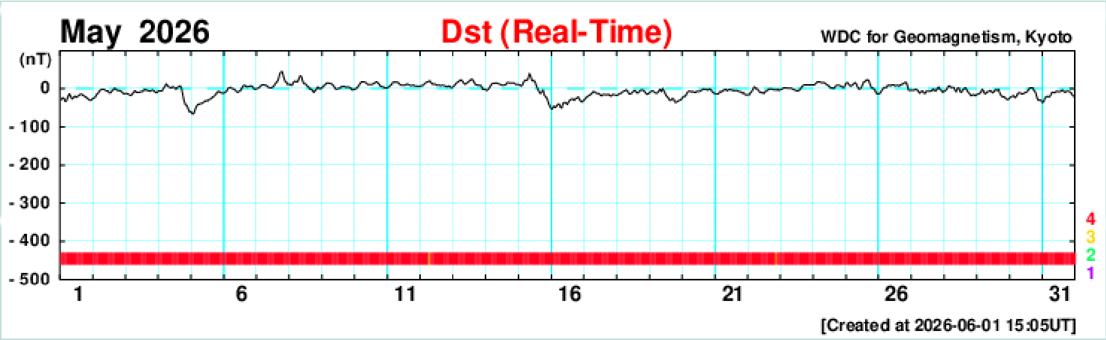
(From – NOAA DSCOVR
In L1, Lagrange Point)



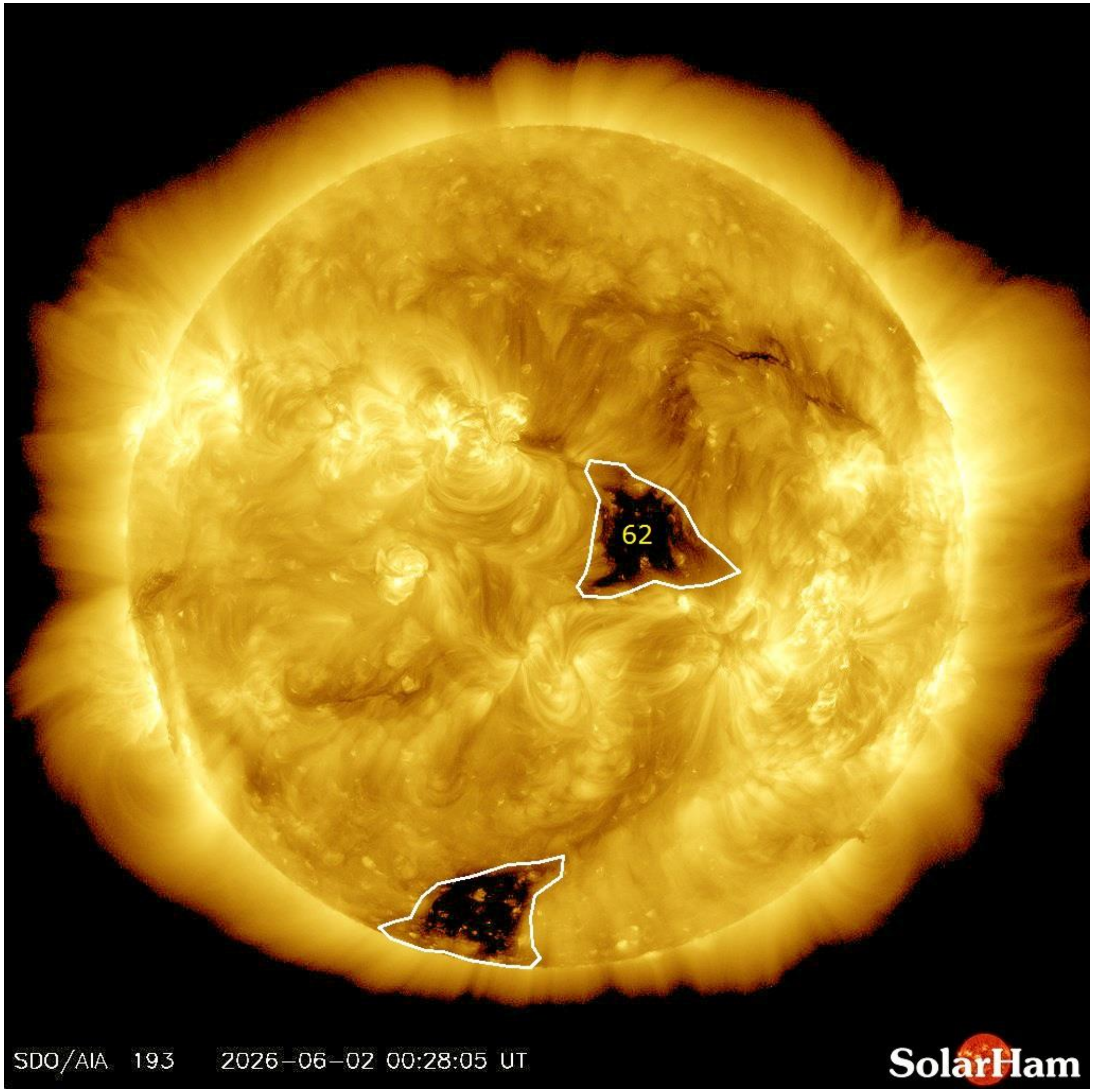
Dst = -6 nT (Ring Field)

(From – Data Analysis Center For Geomagnetics and Space Magnetism – Kyoto University)

From – GOES 18, 19
In geostationary orbit



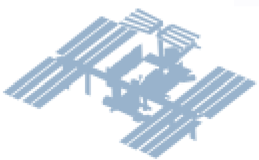
Coronal Holes – 2 JUN 2026

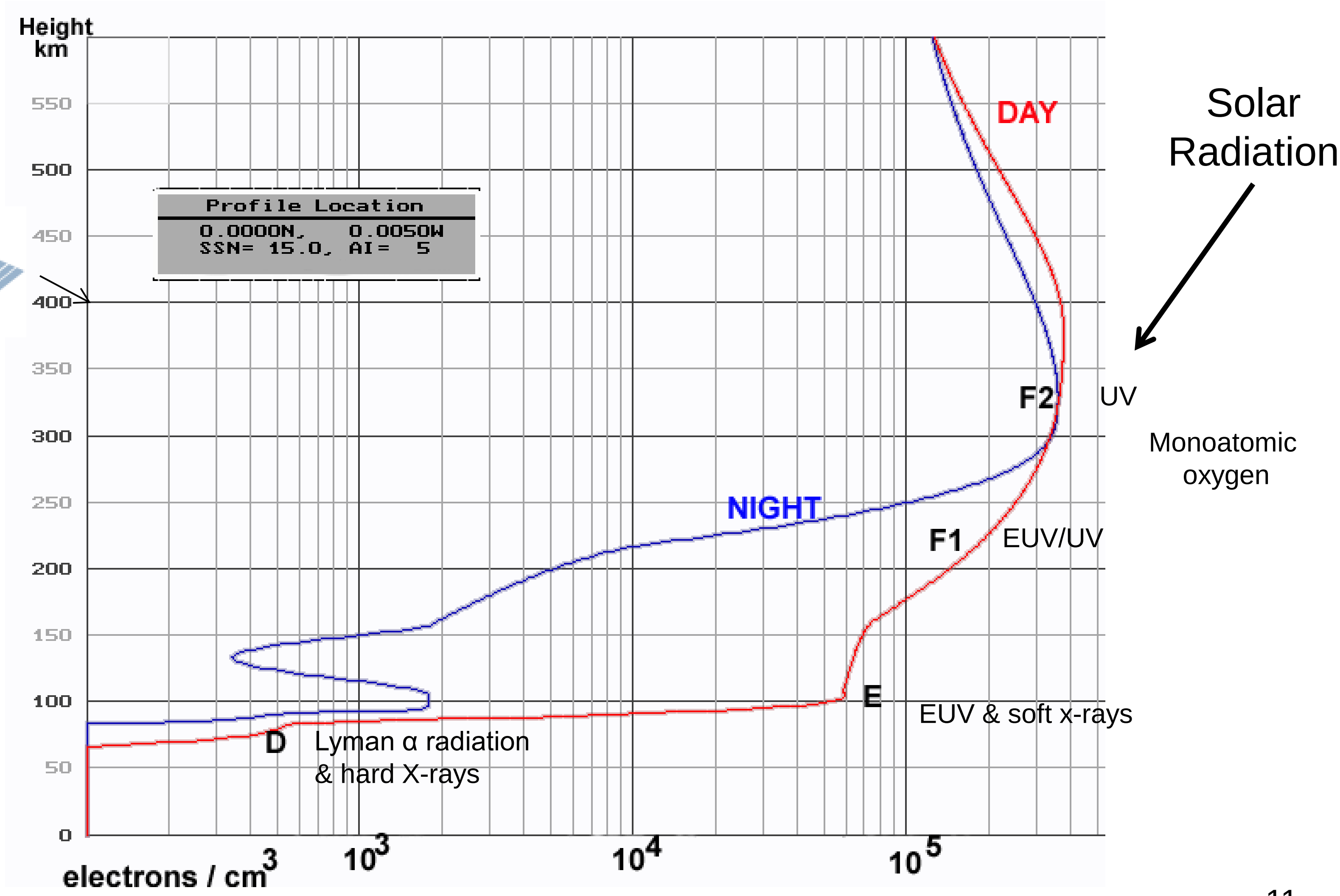


Analysis

Small coronal hole #62 is now facing Earth. A solar wind speed flowing from this zone should reach by June 3rd and into the 4th.

Ionosphere Creation



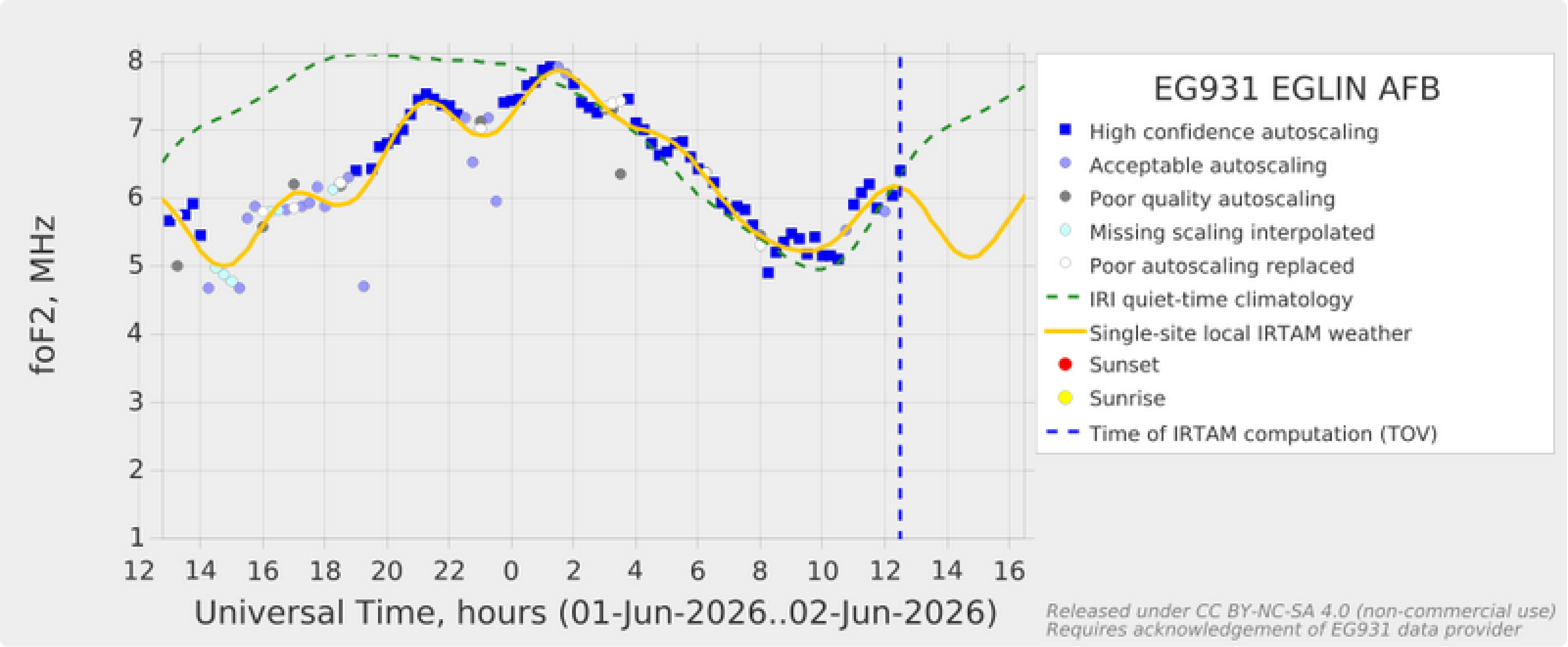


Austin Ionosonde Status

- GIRO is presently doing a software rebuild but GAMBIT is working for Eglin AFB Ionosonde. Austin Ionosonde was heard sounding at my QTH, but Austin GAMBIT plots are modeled only, but appear to be accurate, based on Eglin plot and MARS net performance (Critical Frequency).

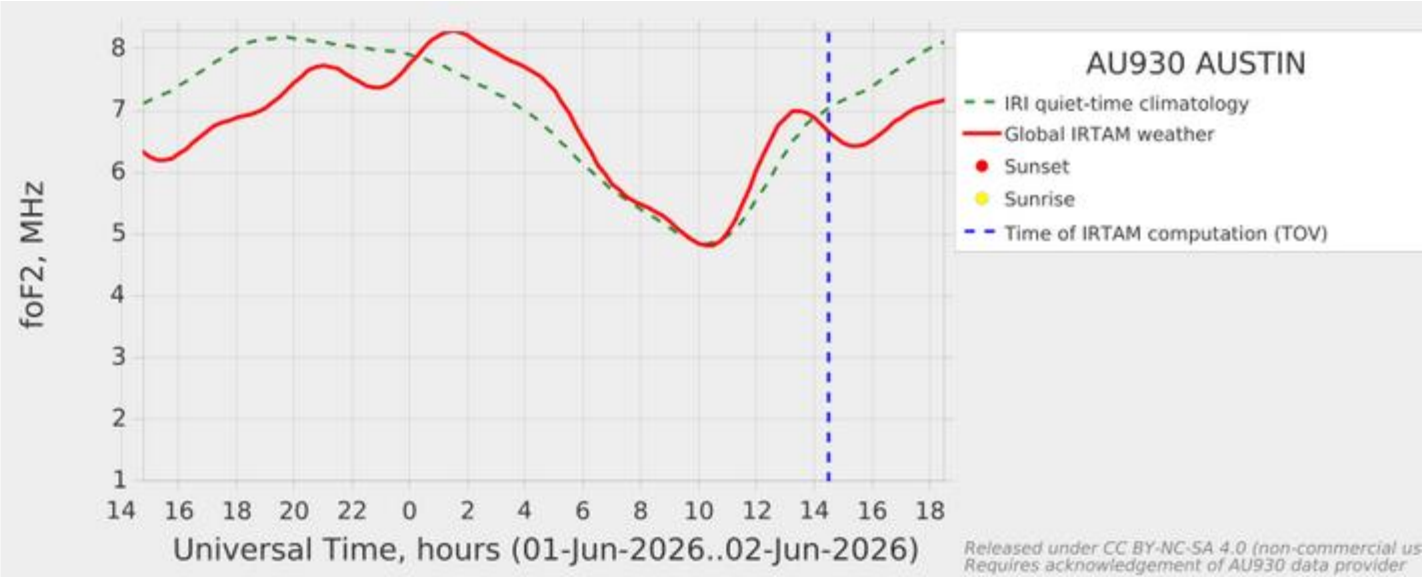
GAMBIT foF2 Trending Chart for Eglin Ionosonde

https://www.region6armymars.org/resources/solarweather.php



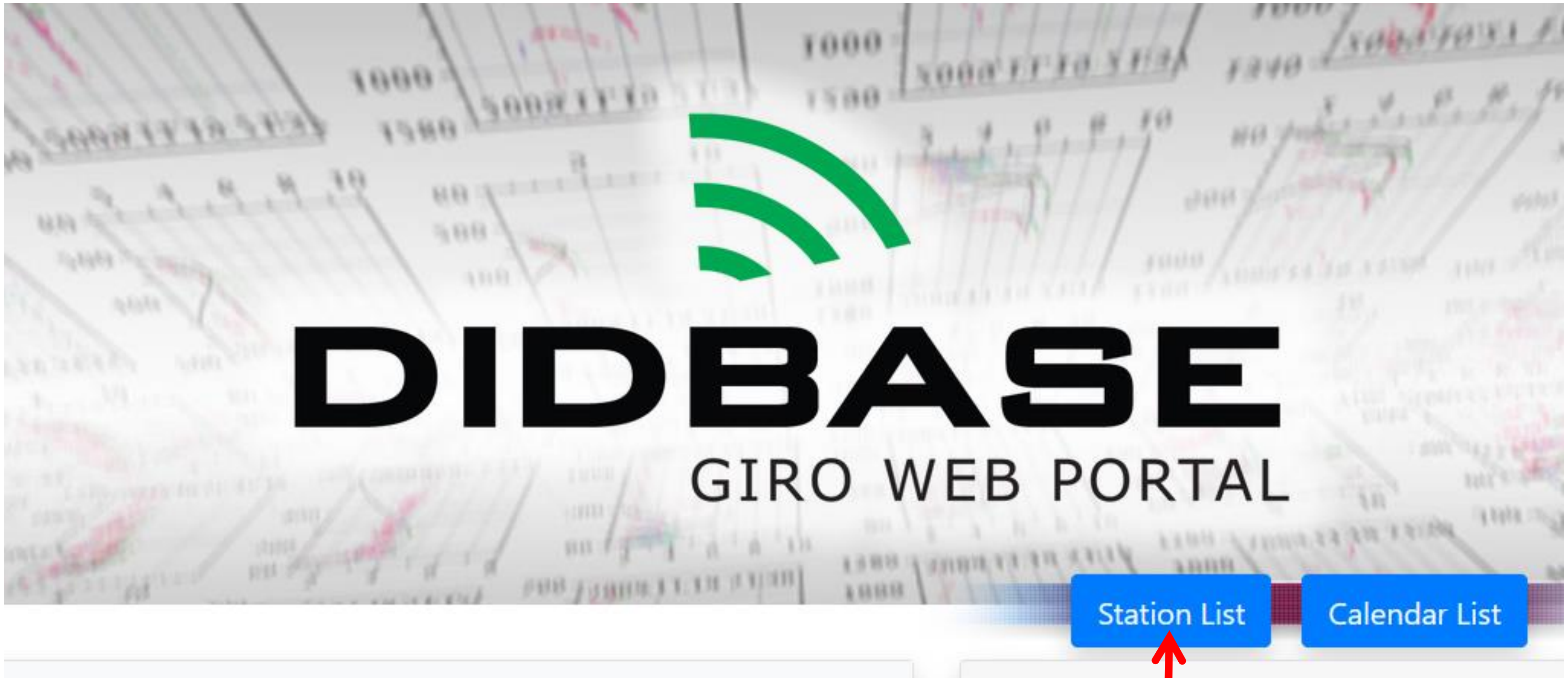
GAMBIT foF2 Trending Chart for Austin Ionosonde – Model only

https://www.region6armymars.org/resources/solarweather.php



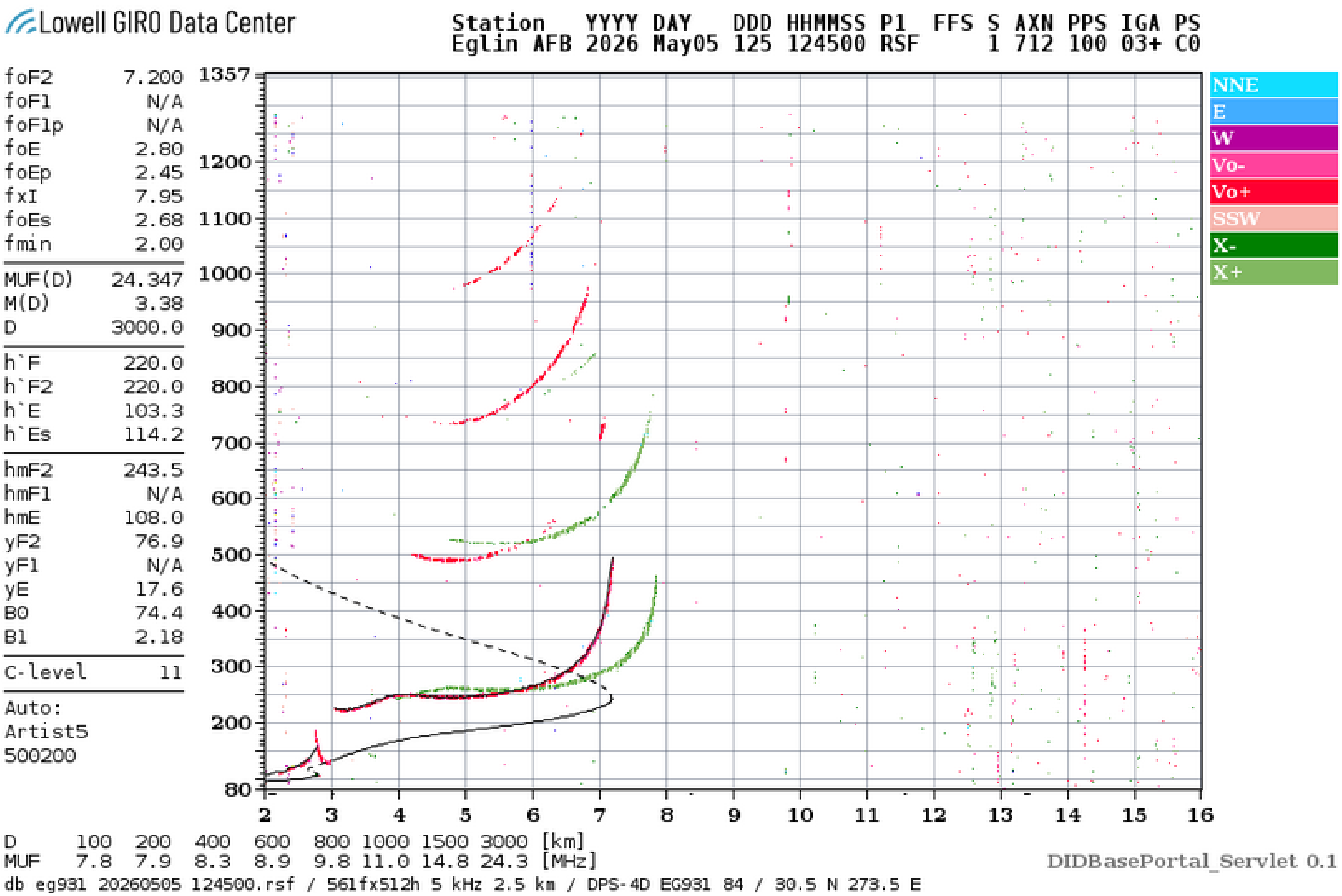
Use of GIRO DIDBASE

https://giro.uml.edu/didbase/



Eglin AFB

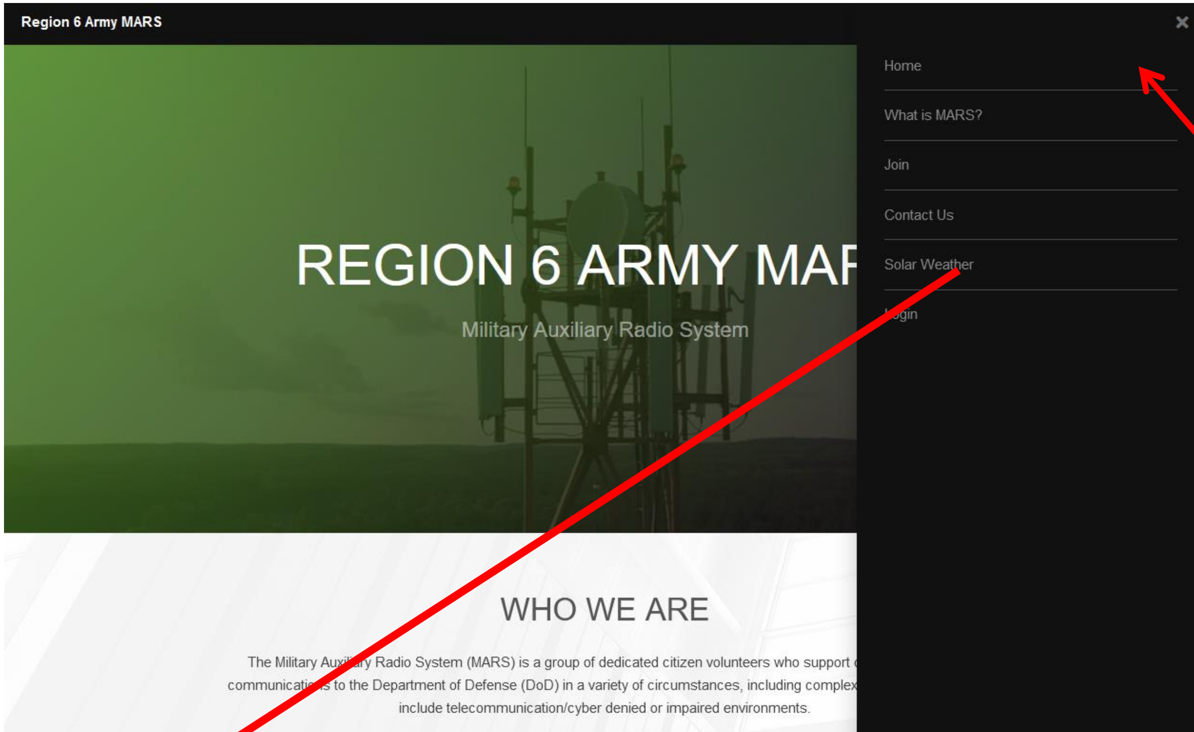
Eglin Ionosonde – 0752 DST (Last Month)



Time Shift for Eglin Ionosonde

- Eglin AFB Ionosonde is east of Austin at about the same Latitude so sun angle is about the same (same solar insolation).
- What happens to the Ionosphere over Eglin will occur over Austin about 45 minutes later.
- This is simply a Longitude difference converted to time.
- So look at Eglin’s foF2 45 minutes earlier to see what is going to happen over Austin.
- The most rapid change in foF2 occurs as the <u>sun is rising and setting (morning and evening).</u>

Solar Weather Data



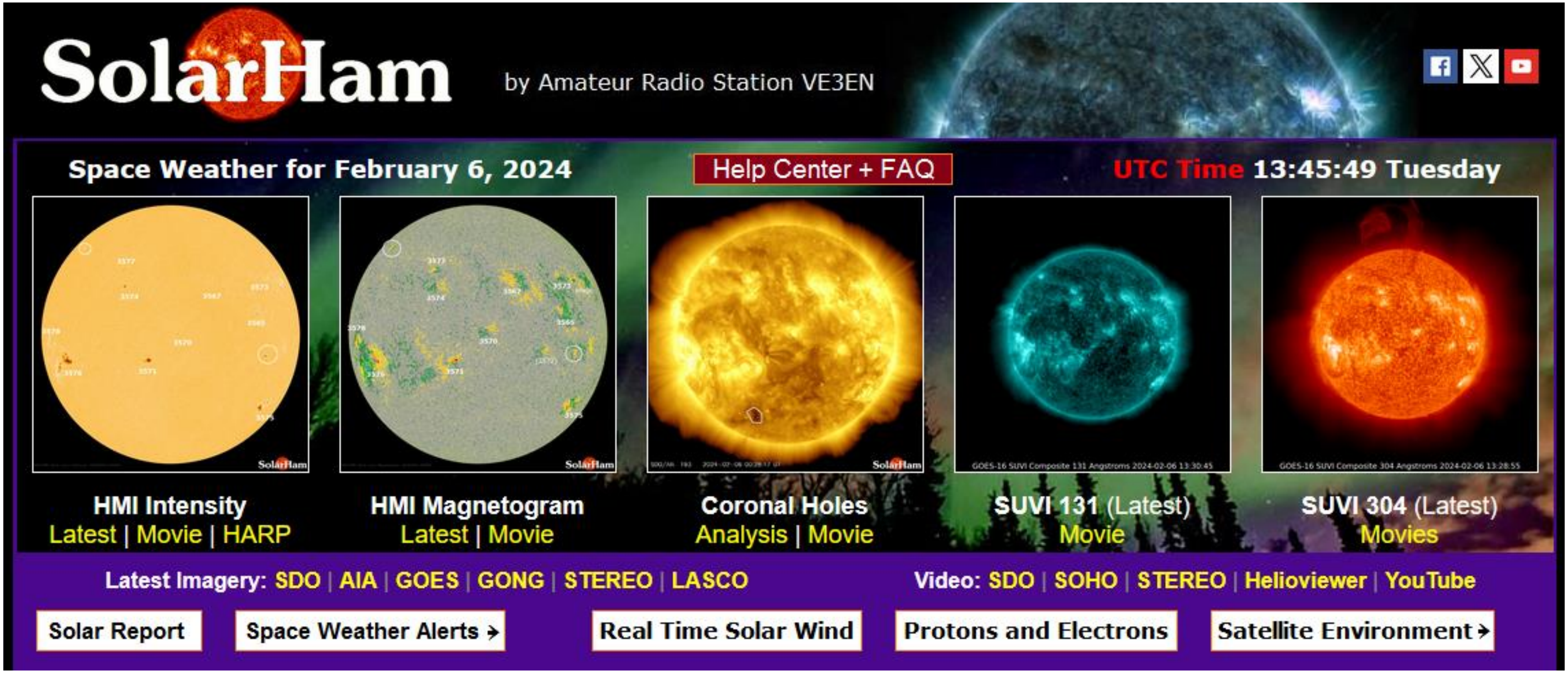
Menu

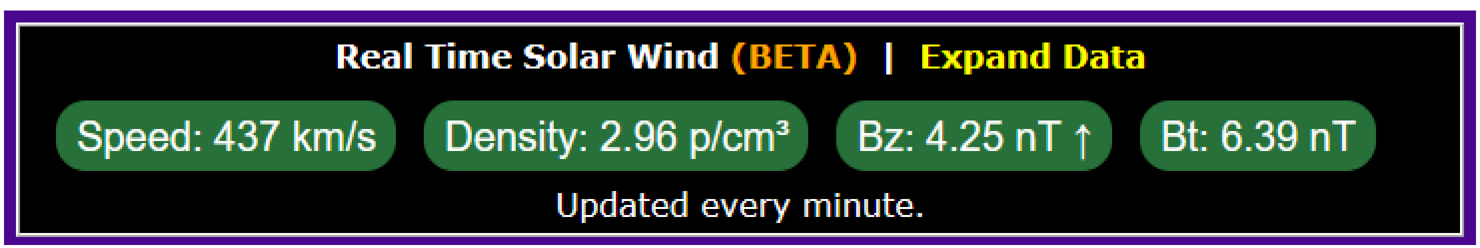
Solar Weather

All Ionosondes
GAMBIT URL

- GAMBIT - Global Assimilative Model of Bottomside Ionosphere Timeline
  - Boulder
  - Eglin

- NOAA Solar Weather - Solar Weather plots of Kp and X-Ray and other solar emissions.
- Solen Solar Weather - Good general solar forecast from an individual.
- Solar Ham - SolarHam provides real time solar news, as well as consolidated data from various sources.


SolarHam
by Amateur Radio Station VE3EN
Space Weather for February 6, 2024
Help Center + FAQ
UTC Time 13:45:49 Tuesday
HMI Intensity
Latest | Movie | HARP
HMI Magnetogram
Latest | Movie
Coronal Holes
Analysis | Movie
SUVI 131 (Latest)
Movie
SUVI 304 (Latest)
Movies
Latest Imagery: SDO | AIA | GOES | GONG | STEREO | LASCO
Video: SDO | SOHO | STEREO | Helioviewer | YouTube
Solar Report
Space Weather Alerts
Real Time Solar Wind
Protons and Electrons
Satellite Environment


Real Time Solar Wind (BETA) | Expand Data
Speed: 437 km/s
Density: 2.96 p/cm³
Bz: 4.25 nT ↑
Bt: 6.39 nT
Updated every minute.

spaceweather.com

News and information about the Sun-Earth environment

https://www.spaceweather.com/

Current Conditions

Solar wind

speed: **314.8** km/sec

density: **9.9** protons/cm^3

more data: [ACE], [DSCOVR]

Updated: Today at 1225 UT

X-ray Solar Flares

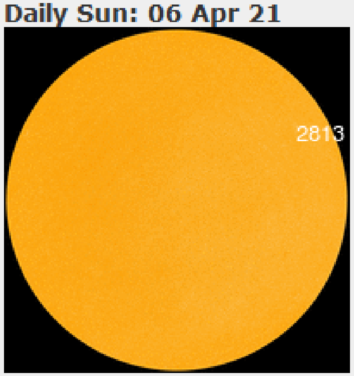
6-hr max: **A1** 1027 UT Apr06

24-hr: **A1** 1515 UT Apr05

explanation | more data

Updated: Today at: 1230 UT

Daily Sun: 06 Apr 21



Sunspot AR2813 is decaying, and poses no threat for strong flares. Credit: SDO/HMI

FLYING TO THE VOLCANO: Iceland's Geldingadalur volcano has turned into an popular tourist attraction---especially since auroras were sighted above the glowing lava. Early this morning, Tuesday, April 6th, Brian Emfinger saw auroras before he even reached the Reykjanes peninsula:

QUESTIONS?

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